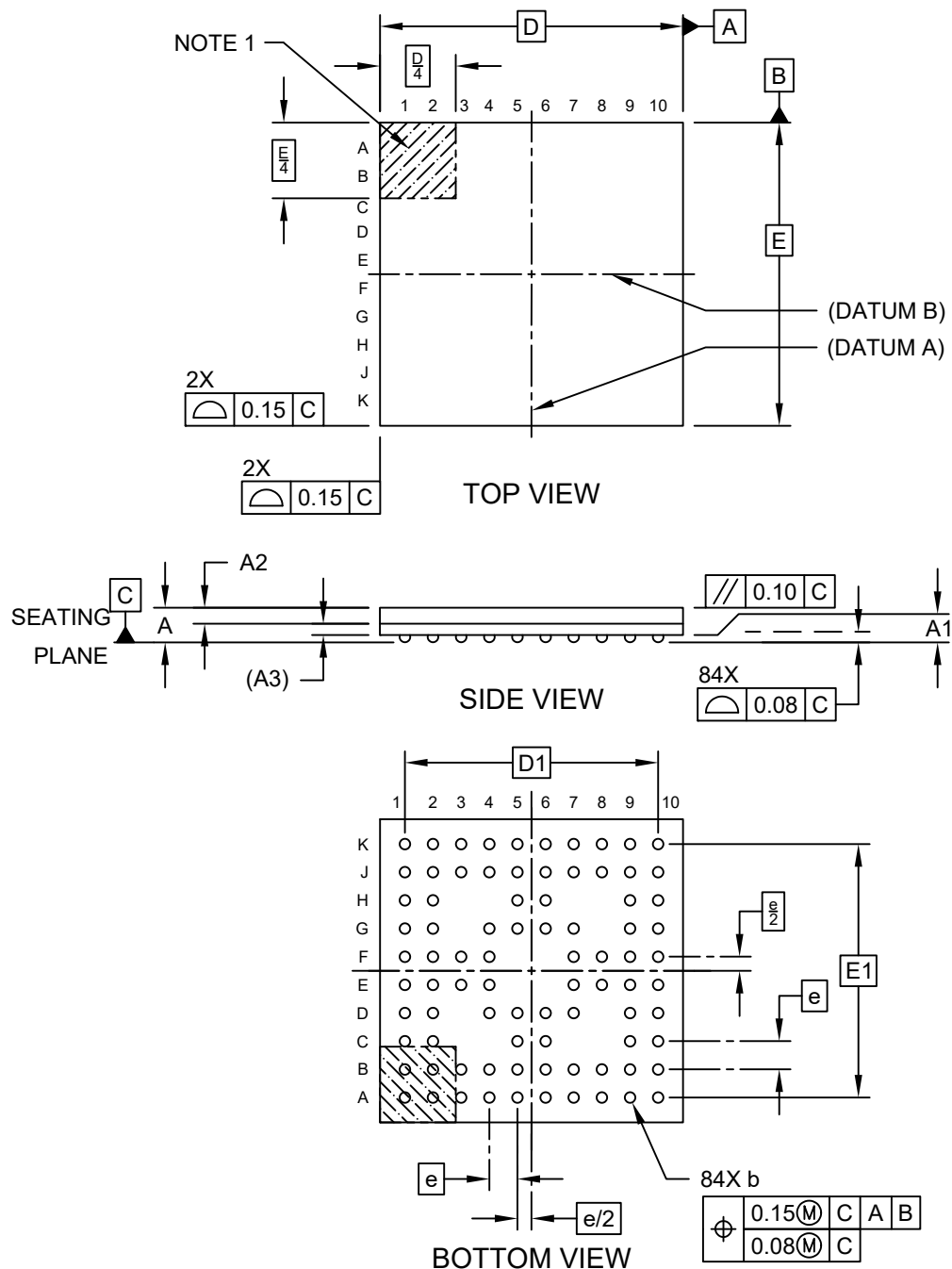


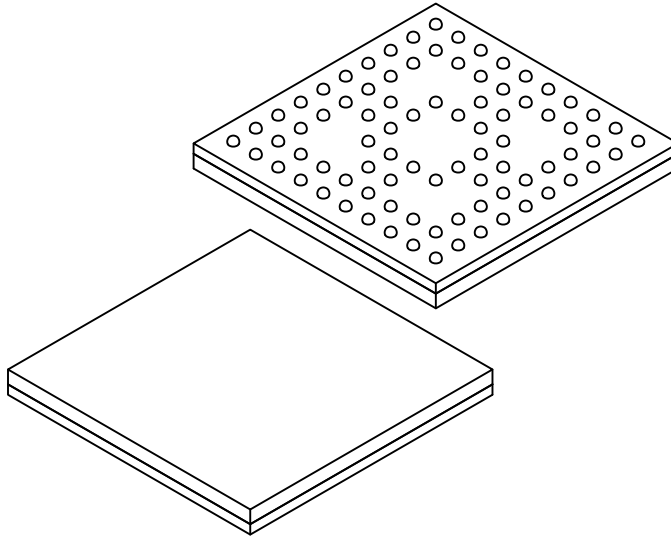
# 84-Ball Very, Very Thin fine Pitch Ball Grid Array (SX, SXX) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 84-Ball Very, Very Thin fine Pitch Ball Grid Array (SX, SXX) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



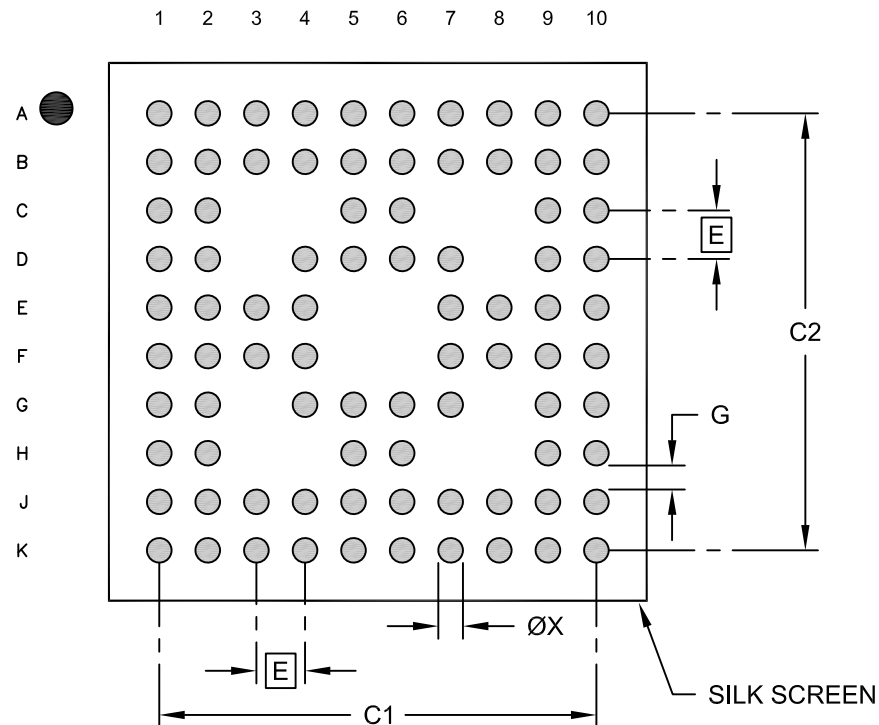
| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Number of Terminals      | N  | 84          |      |      |
| Pitch                    | e  | 0.65 BSC    |      |      |
| Overall Height           | A  | -           | -    | 0.80 |
| Standoff                 | A1 | 0.12        | 0.17 | -    |
| Mold Cap Thickness       | A2 | 0.35        | 0.40 | 0.45 |
| Substrate Thickness      | A3 | 0.13 REF    |      |      |
| Overall Length           | D  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | D1 | 5.85 BSC    |      |      |
| Overall Width            | E  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | E1 | 5.85 BSC    |      |      |
| Ball Diameter            | b  | 0.20        | 0.25 | 0.30 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 84-Ball Very, Very Thin Fine Pitch Ball Grid Array (SX, SSX) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.65 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 5.85 |      |
| Overall Contact Pad Spacing | C2 |             | 5.85 |      |
| Contact Pad Width (X84)     | X1 |             |      | 0.33 |
| Contact Pad to Contact Pad  | G  | 0.25        |      |      |

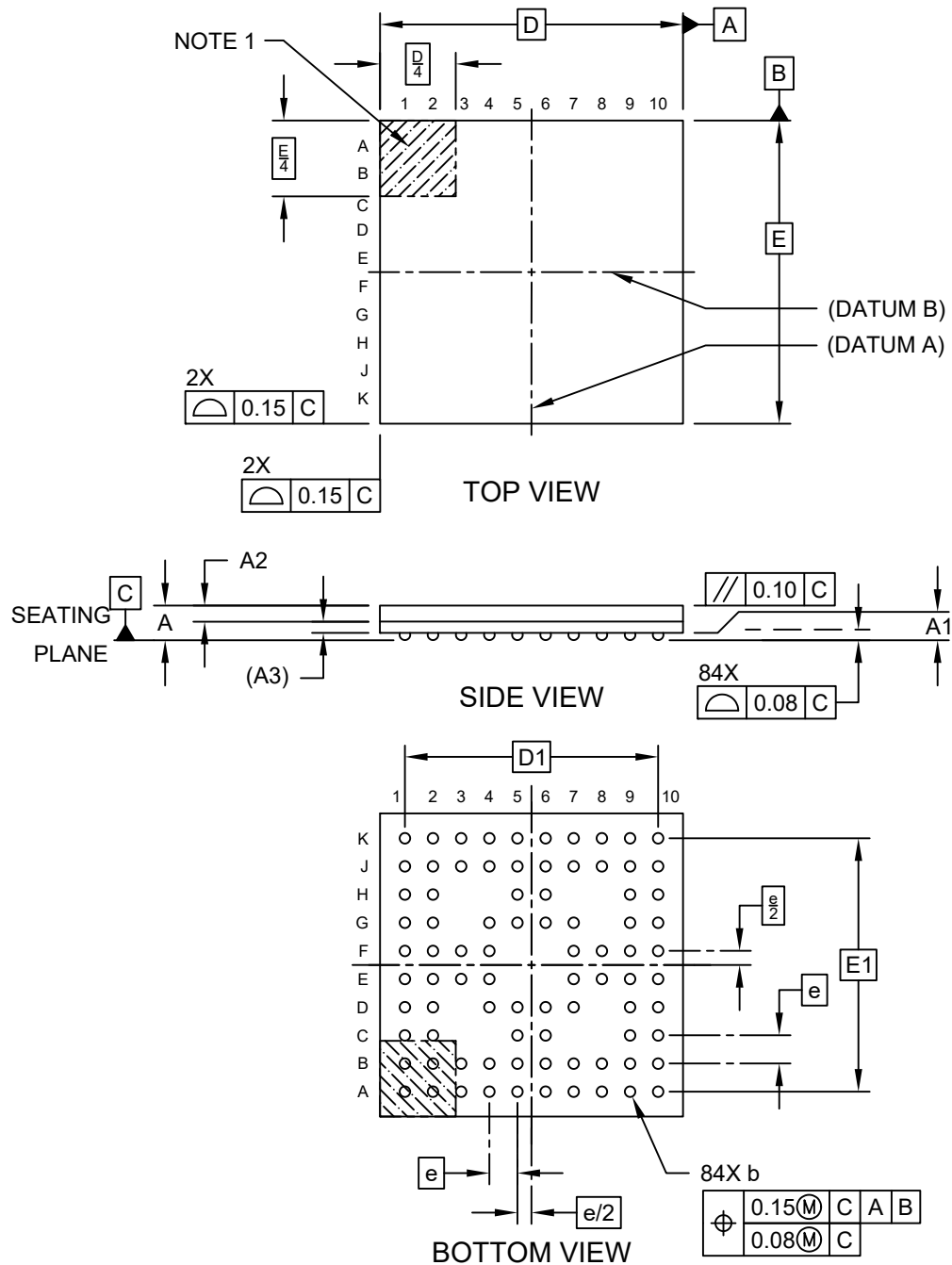
**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

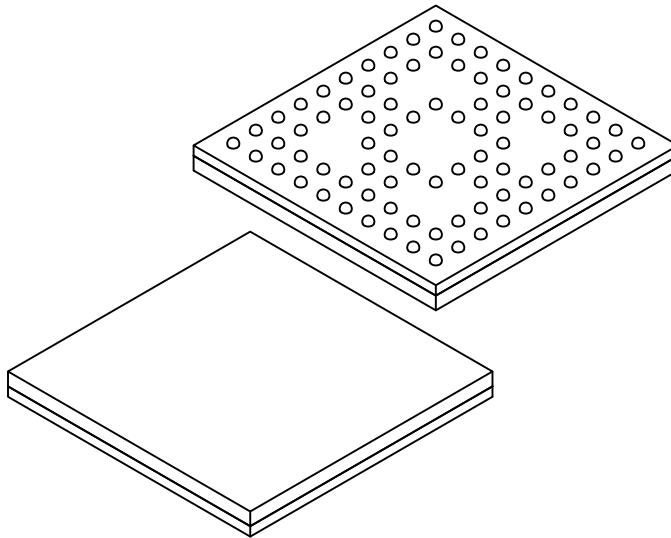
## 84-Ball Very, Very Thin fine Pitch Ball Grid Array (TEX) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 84-Ball Very, Very Thin fine Pitch Ball Grid Array (TEX) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



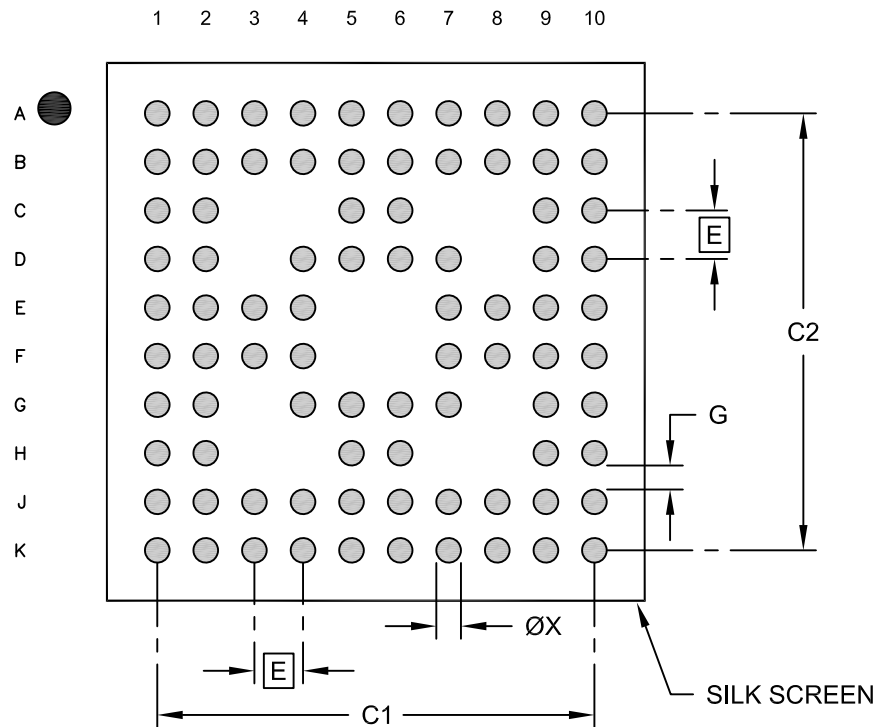
| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Number of Terminals      | N  | 84          |      |      |
| Pitch                    | e  | 0.65 BSC    |      |      |
| Overall Height           | A  | -           | -    | 0.80 |
| Standoff                 | A1 | 0.12        | 0.17 | -    |
| Mold Cap Thickness       | A2 | 0.35        | 0.40 | 0.45 |
| Substrate Thickness      | A3 | 0.13 REF    |      |      |
| Overall Length           | D  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | D1 | 5.85 BSC    |      |      |
| Overall Width            | E  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | E1 | 5.85 BSC    |      |      |
| Ball Diameter            | b  | 0.20        | 0.25 | 0.30 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 84-Ball Very, Very Thin Fine Pitch Ball Grid Array (TEX) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.65 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 5.85 |      |
| Overall Contact Pad Spacing | C2 |             | 5.85 |      |
| Contact Pad Width (X84)     | X1 |             |      | 0.33 |
| Contact Pad to Contact Pad  | G  | 0.25        |      |      |

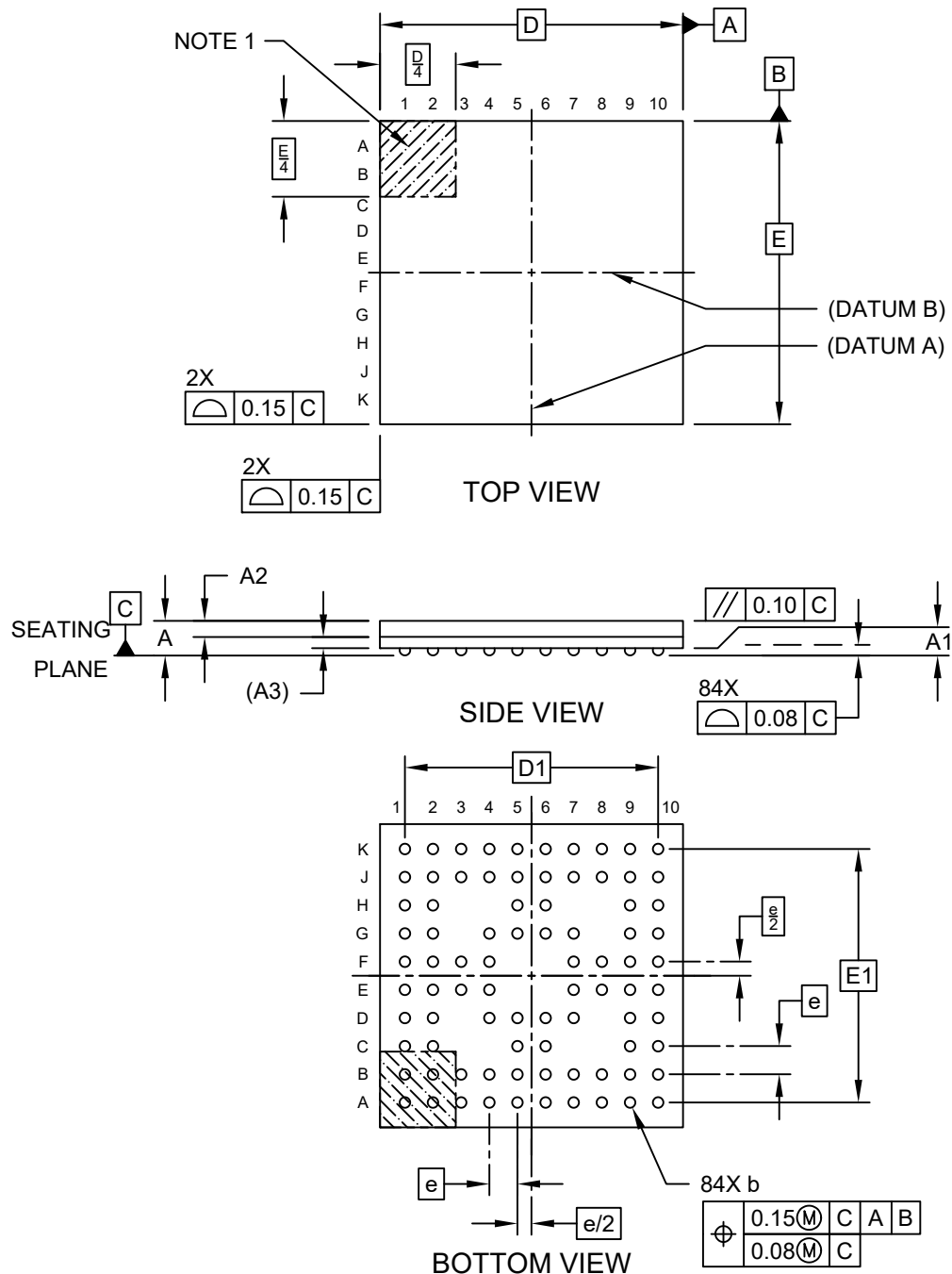
**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

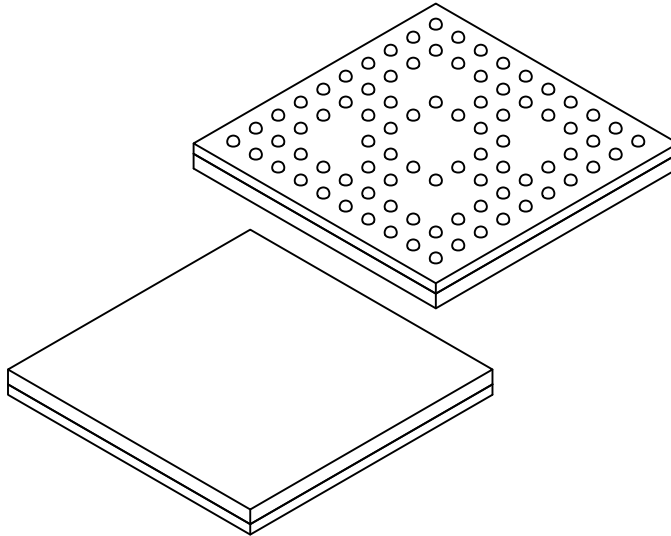
# 84-Ball Very, Very Thin fine Pitch Ball Grid Array (2ZW) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 84-Ball Very, Very Thin fine Pitch Ball Grid Array (2ZW) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Number of Terminals      | N  | 84          |      |      |
| Pitch                    | e  | 0.65 BSC    |      |      |
| Overall Height           | A  | -           | -    | 0.80 |
| Standoff                 | A1 | 0.12        | 0.17 | -    |
| Mold Cap Thickness       | A2 | 0.35        | 0.40 | 0.45 |
| Substrate Thickness      | A3 | 0.13 REF    |      |      |
| Overall Length           | D  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | D1 | 5.85 BSC    |      |      |
| Overall Width            | E  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | E1 | 5.85 BSC    |      |      |
| Ball Diameter            | b  | 0.20        | 0.25 | 0.30 |

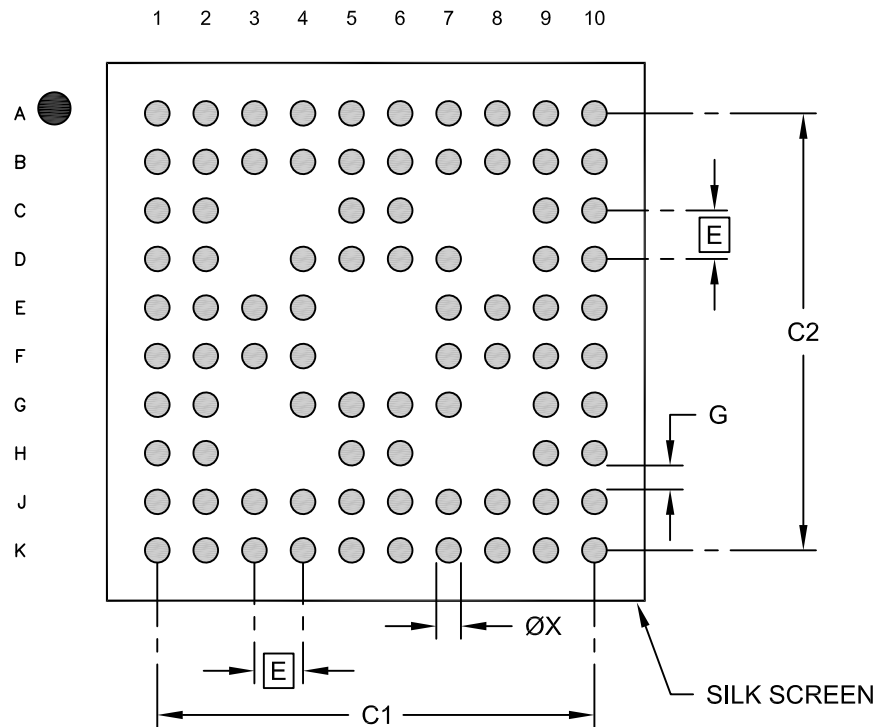
### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.



## 84-Ball Very, Very Thin Fine Pitch Ball Grid Array (2ZW) - 7x7x0.8 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.65 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 5.85 |      |
| Overall Contact Pad Spacing | C2 |             | 5.85 |      |
| Contact Pad Width (X84)     | X1 |             |      | 0.33 |
| Contact Pad to Contact Pad  | G  | 0.25        |      |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.